UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Ensuring the Timely and Orderly
Interconnection of Large Loads

Docket No. RM26-4-000

COMMENTS OF EMERALD AI

Filed: November 21, 2025

I. INTRODUCTION

Emerald AI strongly supports the Federal Energy Regulatory Commission's (Commission) initiative to establish standardized, transparent, and efficient procedures for interconnecting large loads. As an AI-native company that provides an intelligent flexibility management software platform for AI data centers, an "operating system for AI and power," Emerald AI's mission is to enable data centers to flexibly consume energy and serve as valuable grid partners while assuring stringent performance for advanced computing workloads. We aim to enable faster and larger power interconnections for data centers while avoiding and deferring energy infrastructure upgrades, advancing affordability, and bolstering grid reliability.

We are particularly supportive of Principle 7 and urge the Commission to create a "Flexible Load Fast Track" (FLFT) that rewards verifiable flexibility with what large loads value most: faster and larger interconnections for new facilities seeking swift access and existing facilities seeking larger and speedy capacity to upgrade to latest-generation computing technology. The greatest opportunity in this rulemaking is not merely streamlining the administrative study process, but enabling large loads to actively avoid or defer massive grid and energy infrastructure upgrades.

To ensure this fast track is successful, the Commission should adopt a technology-neutral, performance-based definition of flexibility that includes software-defined flexibility as a valid for of "curtailable and dispatchable" load and a substitute for capital-intensive energy infrastructure upgrades that can take years or more. Emerald AI and partners have demonstrated through peer-reviewed scientific results that software-defined data center flexibility is achievable, and along with leading AI and energy partners is on track to deploy these capabilities at commercial scale in the coming months.

II. SUMMARY

Emerald AI respectfully submits these comments in response to the Advance Notice of Proposed Rulemaking (ANOPR) on ensuring the timely and orderly interconnection of large loads.[1]

The convergence of artificial intelligence (AI) and the electric grid represents the defining industrial and infrastructure challenge of the coming decade. As the Federal Energy Regulatory Commission (FERC) initiates rulemaking under Docket No. RM26-4-000 regarding the interconnection of large loads, the energy sector stands at a critical juncture. The traditional "connect and forget" model—where large industrial loads are treated as static, inflexible withdrawals requiring massive firm transmission capacity and peaking generation capacity sized to rare peaks—is fundamentally incompatible with the exponential growth and unique physical characteristics of AI data center demand

Emerald AI provides the intelligent interface between these critical infrastructure networks: the electric grid and AI data centers. Emerald AI was founded in 2024 to solve the exact challenge this ANOPR addresses: the collision between explosive AI-driven load growth and grid capacity limits. This challenge is also the single greatest grid resource opportunity in a generation.

Emerald AI's technology makes AI's power demand flexible, enabling AI data centers to orchestrate their computational workloads and on-site energy resources in response to grid conditions. We agree with the ANOPR's core premise that new rules are needed to manage this growth. Emerald AI's comments focus particularly on **Principle 7**, as we believe its successful implementation of speeding time-to-power for flexible data centers is the key to unlocking AI innovation, enhancing U.S. competitiveness, and ensuring grid reliability.

The economic stakes of this rulemaking are exceptionally high. Analysis indicates that investment in AI infrastructure and data centers is currently a primary driver of U.S. economic expansion, contributing significantly to GDP growth in the first half of 2025. Delays in interconnection, driven by study processes that do not allow for flexibility–including software-defined flexibility–threaten to stall this economic engine. By adopting a technology-neutral, performance-based definition of flexibility and curtailable load, the Commission can unlock tens of gigawatts of capacity, ensuring that the U.S. maintains its competitive edge in AI while protecting ratepayers from the costs of unnecessary transmission buildout.

The traditional approach to interconnecting large loads involves studying the facility at its peak potential withdrawal under severe grid stress conditions. If this analysis reveals a violation of thermal or voltage limits, the interconnecting customer may trigger grid upgrades—new transmission lines, substation expansions, or capacitor banks. These upgrades are capital

---

[1] U.S. Department of Energy, Office of the Secretary, *Secretary of Energy's Direction that the Federal Energy Regulatory Commission Initiate Rulemaking Procedures Regarding the Interconnection of Large Loads Pursuant to Section 403 of the Department of Energy Organization Act*, October 23, 2025.

intensive and slow to deploy. A new high-voltage transmission line can take 7–10 years to permit and build, while a data center can be constructed in 18–24 months. Furthermore, if the costs of these upgrades are deemed prohibitive by the developer, the project may be cancelled, or the costs may be socialized across the rate base if the upgrades provide incidental system benefits.

The “Flexible Load Fast Track” (FLFT) proposal offers a mechanism to break this deadlock. By utilizing software to “shave” the peaks of AI data center demand through computational flexibility and onsite energy resources, or shift that computational demand spatially, grid operators can defer these heavy infrastructure investments. The economic logic is compelling: the marginal cost of software-based flexibility is near-zero (operating expense), whereas the marginal cost of new transmission capacity is extremely high (capital expense).

III. DEFINING FLEXIBILITY

Principle 7 calls for expediting studies for “curtailable and dispatchable” loads. The Commission's definition of this term will determine the success or failure of the FLFT.

Historically, utilities and NERC standards have often implicitly or explicitly defined “curtailable load” as a load that can be physically disconnected via a utility-controlled relay or circuit breaker. While highly reliable, this is a blunt instrument unsuitable for modern digital infrastructure. For a data center, a total power cut is catastrophic—it risks damaging millions of dollars of hardware, corrupting data, and violating customer Service Level Agreements (SLAs). Consequently, data centers typically reject “interruptible” tariffs that rely on breaker trips. The Commission should instead adopt a performance-based definition of flexibility, consistent with the shift toward performance-based standards seen in other areas of grid regulation. If a facility can demonstrate, through telemetry and secure software interfaces, that it will reduce its withdrawal from 100 MW to 75 MW within the required timeframe of a signal, it effectively provides the same reliability value as a physical breaker, but with vastly higher economic efficiency. Modern AI data centers possess sophisticated, inherent flexibility that is controllable through software.

The final rule should be technology-neutral and performance-based, recognizing the full spectrum of on-site flexibility. Emerald AI recommends the Commission explicitly recognize the following forms of data center flexibility as valid methods of curtailment, all of which can be orchestrated by a facility's flexibility management platform:

1. **Temporal Flexibility:** The ability of software to intelligently and briefly slow, pause, or reschedule batchable, non-time-sensitive AI workloads (such as fine-tuning or batch inference such as synthetic data generation) in response to a grid signal, all while assuring stringent AI performance requirements.

2. **Spatial Flexibility:** The ability of software to shift workloads to alternate facilities in regions connected by high-speed fiber networks to follow the availability of power and avoid local grid constraints.
3. **Resource Flexibility:** The ability of software to co-optimize the two methods above alongside the dispatch of on-site resources, such as batteries, generators, thermal storage, cooling operations, fuel cells, or other on-site resources and DERs, for a maximum, reliable, and unified response.

In addition, a fourth type: adjacent or offsite flexibility, deserves further study to complement the other three types of data center flexibility to complement and augment the total flexibility a facility can offer. Such adjacent flexibility refers to the ability to coordinate with nearby or aggregated resources (e.g., a local VPP) via a software interface, so that the net effect reduces withdrawals on constrained parts of the grid.

By focusing on measurable, verifiable performance, the Commission will enable innovation in AI infrastructure rather than prescribing how data centers must provide flexibility. Software-enabled flexibility, such as Emerald's platform, can unlock gigawatts of capacity on today's grid by controlling when and where power is consumed, rather than relying solely on "steel-in-the-ground" upgrades. Under such a technology-neutral framework, a data center may use a software-only solution, a hardware-only solution, or a hybrid of the two to meet its FLFT commitment.

A technology-neutral framework is so important because of the fundamentally transformative potential of AI data centers to be flexible power consumers. AI data centers are unique compared with other energy users. They can be massive point sources of load that can materially provide relief through flexibility; they respond swiftly and are virtually controllable; and they are connected at the speed of light to other data centers. Unlike a factory that must consume power where it is physically located, AI workloads can be migrated between data centers connected by high-speed fiber optics. If the PJM interconnection is thermally constrained but the MISO region has excess wind generation, an orchestrator can route new inference requests to the MISO facility, for example. This capability acts as a "virtual transmission line." Instead of moving megawatts of electrons across a congested wire to meet a static demand, the system moves the demand for those electrons to a location where power is abundant. This alleviates local congestion without the need to break ground on new transmission towers, offering a powerful "Non-Wires Alternative" (NWA) consistent with FERC Order 2023 objectives.

Emerald AI's platform operationalizes this model. It sits between grid signals and AI workload schedulers, translates system conditions into power targets, and orchestrates AI workloads and on-site resources to deliver a pre-committed flexibility response. In practice, an Emerald AI demonstration with partners reduced AI workload power consumption by 25% over three hours during a grid stress event while fully preserving compute service quality.[2] The results of this collaboration have been peer-reviewed and accepted for publication in one of the world's top scientific journals. This real-world performance illustrates the type of "curtailable" behavior the Commission should recognize under Principle 7. In addition, Emerald AI and its partners have announced the Aurora project in Virginia as the "first power-flexible AI Factory" operating at commercial scale, 96MW, as a scaled proof point of the capabilities of software-defined flexibility.[3]

With respect to process design, Emerald AI recommends that transmission providers establish a dedicated Flexible Load Fast Track process, parallel to the existing serial queue. The FLFT would allow facilities meeting defined flexibility thresholds to receive interconnection approval within 60 to 90 days, based on verification of available firm capacity and demonstration of control and telemetry systems.

IV. SECURE SOFTWARE INTERFACE

Principle 7 rightly requires that the system operator's "ability to control" a flexible facility must be sufficient. Emerald AI strongly recommends this be defined as a secure, standardized, and auditable software interface, not direct, physical utility control.

This model provides the proper balance of reliability and security:

1. **Signal:** The system operator sends a high-level dispatch signal (e.g., "Curtail by this many MW") to the facility's registered flexibility management platform.
2. **Orchestration:** The facility flexibility management software orchestrates computational flexibility and energy resources such as batteries, cooling, generators, and fuel cells to meet the MW target without compromising its mission-critical operations.
3. **Verification:** The platform provides real-time, auditable Measurement & Verification (M&V) data back to the operator to confirm compliance.

This "intelligent interface" model is a robust approach that provides system operators with the verifiable, fast-acting resource they need, while giving data centers the operational autonomy and cybersecurity they require to manage mission-critical, complex AI workloads.

V. ALIGNMENT WITH OTHER ANOPR PRINCIPLES

Emerald AI's recommendation for a performance-based Flexible Load Fast Track (FLFT) can be a key mechanism for implementing several of the Commission's other principles in a cohesive, market-based manner.

---

[2] NVIDIA, *AI Factories Flex Their Power Use to Help Balance the Grid*, October 2025.
[3] Axios, NVIDIA's *Aurora AI Center Shows How Flexible Power Use Could Reshape Electric Bills*, October 29, 2025.

- Principle 5 (Hybrid Facilities): AI data centers increasingly pair large loads with on-site generation or storage. Emerald AI's platform treats these as a single hybrid system, co-optimizing computational flexibility and on-site assets to produce a predictable net withdrawal profile. A facility that commits a MW-capacity of verifiable flexibility through these various options can be studied and interconnected based on that net firm withdrawal, directly advancing Principle 5's goal of appropriately treating hybrid large-load projects.

- Principle 8 (Cost Allocation): The FLFT, enabled by verifiable software-defined flexibility, creates a clear economic choice: projects that remain inflexible proceed through the standard process and may bear upgrade-related costs, whereas projects that commit to and deliver flexibility can avoid or reduce those upgrades and connect sooner. Emerald's platform makes this choice feasible by turning flexibility into a controllable, measurable product. This ensures that upgrade costs are not unjustly socialized onto other ratepayers when a load can mitigate its own impact through performance.

VI. CRITICAL GAPS IN EXISTING TRANSMISSION SERVICE AND PLANNING FOR LARGE FLEXIBLE LOADS

Despite growing interest in large flexible loads–including AI data centers capable of providing verifiable, performance-based curtailment–no transmission service framework currently exists in the United States that allows such loads to avoid or mitigate network upgrades through demonstrated flexibility.

This gap means that even highly flexible loads—capable of reliably reducing or shifting hundreds of megawatts—are studied as if they were inflexible, static, worst-case withdrawals. As recent analysis by Nicholas Institute and Roselle explains: "Transmission service frameworks do not yet exist to allow for load flexibility to avoid network upgrades. Network Integration Transmission Service at the wholesale level is non-interruptible under FERC's pro forma tariff. And, while non-firm point-to-point service does exist, there is no non-firm network option for a load to access and draw upon the full network during times it is not curtailed. Nor do utilities studying large load interconnection generally provide a pathway through which a curtailable customer can be studied to avoid network upgrades." [4] The result is an overestimation of required network upgrades, longer study timelines, and significant cost assignments that do not reflect the resource's actual operational profile.

Principle 7 is the Commission's opportunity to correct this structural deficiency. By establishing a Flexible Load Fast Track (FLFT) that recognizes verifiable, software-enabled curtailment as a serviceable substitute for firm transmission, the Commission can provide a pathway for flexible

[4] Nicholas Institute for Energy, Environment & Sustainability, *How DOE's Proposed Large Load Interconnection Process Could Unlock the Benefits of Load Flexibility*, October 2025

loads to interconnect without triggering unnecessary upgrades, thereby enhancing grid reliability and reducing ratepayer costs.

Today, transmission providers conduct generator interconnection planning and large-load interconnection planning in parallel but largely disconnected processes. This separation is increasingly incompatible with the realities of modern grid operations, where large AI loads may be capable of providing fast, measurable, dispatchable flexibility on par with many supply-side resources.

Absent a unified planning framework:

- Transmission providers evaluate large loads as rigid, “always-on” withdrawals.
- They do not evaluate whether flexible load behavior could mitigate or eliminate the need for upgrades.
- They do not co-optimize load flexibility and generation additions within a shared regional planning model.

This siloed approach can lead to suboptimal upgrade decisions, causing transmission providers to pursue unnecessarily expensive network upgrades that could be avoided with a more integrated planning architecture. In some regions, this fragmentation has led to upgrade determinations that are larger and more costly than needed, raising potential concerns about whether such costs are “just and reasonable” under the Federal Power Act.

A performance-based FLFT directly addresses this issue. By incorporating verifiable load flexibility into interconnection studies, transmission providers can evaluate net impacts rather than worst-case, inflexible scenarios. This results in:

- More accurate assessments of needed upgrades
- More efficient regional planning
- Lower cost burdens for all transmission customers
- Greater utilization of existing grid capacity

In short, integrating flexible-load capabilities into planning processes is essential to preventing overbuilding, accelerating interconnections, and ensuring that rates remain just and reasonable.

VII. CATEGORIES OF PERFORMANCE ATTRIBUTES FOR FLEXIBLE LOADS

To implement a flexible-load participation framework in a manner that is both fair and durable, the Commission may wish to articulate clear categories of performance standards that transmission providers should consider when evaluating flexible-load commitments. These standards help ensure that flexibility is credited appropriately, integrated consistently across regions, and protected from misuse, while still allowing technology-neutral innovation.

Emerald AI respectfully suggests the following six categories of performance criteria the Commission could reference as reasonable guardrails for flexible-load participation:

1. Event Duration Standards: Facilities should specify the maximum duration for which they can sustain curtailment during a flexibility event. Establishing predictable duration parameters enables transmission providers to rely on the resource in planning and operational studies and bound the commitment from large loads to provide curtailment in exchange for faster and larger power connections.

2. Advance Notice Requirements: Reasonable notice parameters (e.g., minimum dispatch windows) ensure that flexible loads can prepare their workloads and internal systems to deliver a reliable response without compromising mission-critical operations.

3. Recovery Period Protection: Following a curtailment event, flexible facilities may require a defined recovery period to return to normal operations without creating secondary load spikes. Clear standards prevent operators from over-calling events in ways that undermine reliability benefits.

4. Annual Event Limits: Setting a reasonable ceiling on annual curtailment events prevents excessive dispatch and ensures that flexible loads are used strategically, during genuine system stress, rather than as routine operational tools. AI data centers and next-generation AI Factories engage in the most economically critical workloads of the entire economy, so their curtailment should be extremely judicious.

5. Response Time and Tolerance Limits: A performance-based framework should include measurable expectations regarding the speed and accuracy with which a facility must reduce load after receiving a dispatch signal. These limits provide operators with clarity and ensure that flexible loads deliver consistent, verifiable value.

6. Testing and Compliance Rules: Periodic testing and ongoing monitoring are essential to ensure the reliability of flexible-load commitments over time. Testing frameworks should be technology-neutral and recognize software-based workload orchestration as a valid and auditable means of demonstrating performance.

Together, these categories offer a balanced structure: they ensure that flexible loads are appropriately credited for the real reliability value they provide, while also establishing safeguards that maintain system integrity and prevent over-reliance or misuse. Importantly, these standards can be implemented without prescribing any specific technology or operating model, allowing innovation to continue while giving the Commission and transmission providers the certainty they require.

VIII. CONCLUSION

Emerald AI recognizes the Commission for this timely and critical proceeding. By establishing a Flexible Load Fast Track anchored in a technology-neutral and performance-based definition of flexibility, the Commission can turn AI's new power demand into the grid's newest and most valuable resource. Emerald AI respectfully urges the Commission to finalize a rule that ensures software-defined flexibility is a valid pathway to achieve faster and larger interconnections, thereby accelerating AI innovation, strengthening the grid, and protecting affordability for all consumers. We look forward to working with the Commission as it moves through this rulemaking process.

Respectfully submitted,
Dr. Varun Sivaram
Chief Executive Officer
Phone: +1(408)-656-0083
Email: varun@emeraldai.co

Emerald AI
Washington, DC
November 21, 2025